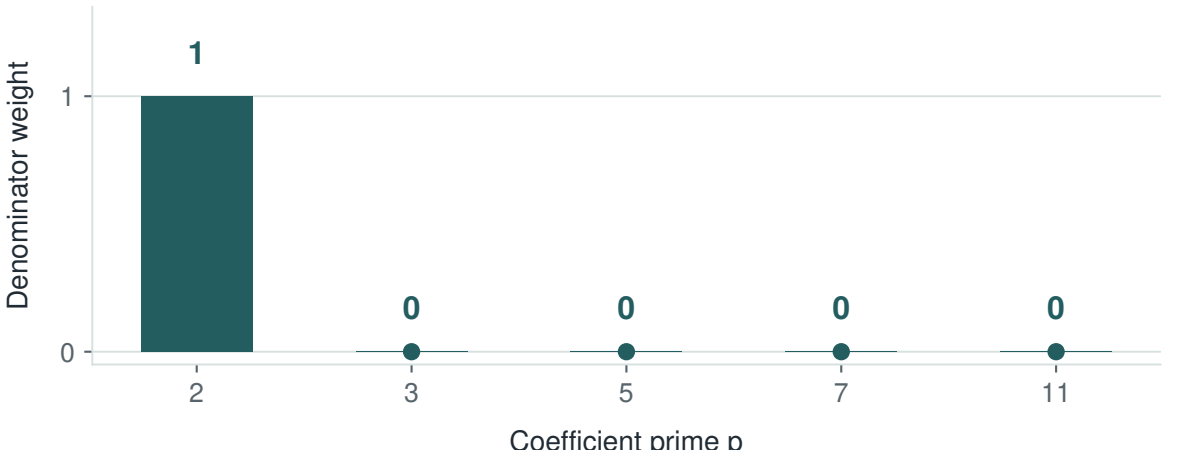
The denominator singles out the prime two

The exact local weight of each sign projector is $v_p(2)$.



The denominator 2 is a unit at every odd prime.